$P(H_2) = 0.4 \quad \Rightarrow P(\bar{H}_2) = 1 - P(H_2) = 0.6$

$P(H_3) = 0.2 \quad \Rightarrow P(\bar{H}_3) = 1 - P(H_3) = 0.8$

$P(H_4) = 0.1 \quad \Rightarrow P(\bar{H}_4) = 1 - P(H_4) = 0.9$

The probability that the gun hits the plane is

$$P(H) = 1 - P(\bar{H}) = 1 - P(\overline{H_1 \cup H_2 \cup H_3 \cup H_4})$$

$$= 1 - P\left(\bar{H}_1 \cap \bar{H}_2 \cap \bar{H}_3 \cap \bar{H}_4\right)$$

$$= 1 - P(\bar{H}_1)P(\bar{H}_2)P(\bar{H}_3)P(\bar{H}_4)$$

$$= 1 - (0.8)(0.6)(0.8)(0.9) = 1 - 0.3456$$

$$P(H) = 0.6544$$

Example 12.22

X speaks truth in 70 percent of cases, and Y in 90 percent of cases. What is the probability that they likely to contradict each other in stating the same fact?

Solution

Let A be the event of X speaks the truth, B be the event of Y speaks the truth

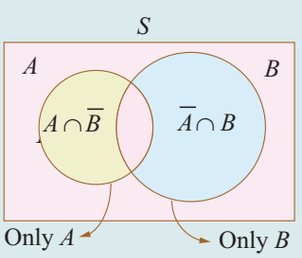
$\therefore \bar{A}$ is the event of X not speaking the truth and $\bar{B}$ is the event of Y not speaking the truth.

Let C be the event that they will contradict each other.

Given that

$P(A) = 0.70 \quad \Rightarrow P(\bar{A}) = 1 - P(A) = 0.30$

$P(B) = 0.90 \quad \Rightarrow P(\bar{B}) = 1 - P(B) = 0.10$



C = (A speaks truth and B does not speak truth or B speaks truth and A does not speak truth)

$C = [(A \cap \bar{B}) \cup (\bar{A} \cap B)]$ (see figure)

since $(A \cap \bar{B})$ and $(\bar{A} \cap B)$ are mutually exclusive,

$$P(C) = P(A \cap \bar{B}) + P(\bar{A} \cap B)$$

$$= P(A)\,P(\bar{B}) + P(\bar{A})\,P(B)$$

(Since A, B are independent event, A, $\bar{B}$ are also independent events)

$$= (0.70)\ (0.10) + (0.30)\ (0.90)$$

$$= 0.070 + 0.270 = 0.34$$

$$P(C) = 0.34.$$

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